

SECURITY INCIDENT REPORT

SOC Home Lab
Attack Detection & Analysis

Analyst	Date	Classification	Status
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Executive Summary

A controlled penetration test was conducted against an intentionally vulnerable target system (Metasploitable 3) within an isolated VirtualBox virtual network. Three attack scenarios were executed and monitored in real time using Splunk Enterprise SIEM. All three attacks were successfully detected and logged, producing a total of 1,961 security events. Critical findings include an excessive attack surface with 9 exposed services, default credentials on SSH, and a remote code execution vulnerability in ProFTPD 1.3.5 that allowed full system compromise.

1,961	3 / 3	1,830+	CRITICAL
Total Events Captured	Attacks Detected	Events at Peak Hour	Highest Severity

Lab Environment

Component	Details	IP Address
Attacker	Kali Linux 2026.1	10.0.2.4
Victim	Metasploitable 3 (Ubuntu 14.04)	10.0.2.15
SIEM	Splunk Enterprise (Windows Host)	192.168.56.1
Network	VirtualBox NAT Network + Host-Only Adapter	Isolated
Log Forwarding	rsyslog → UDP 514 → Splunk port 514	—

Findings

FINDING 1 — Network Reconnaissance (Port Scan) SEVERITY: HIGH

Tool	Nmap 7.98
Command	nmap -sS -sV 10.0.2.15
Duration	14.19 seconds
Ports Discovered	9 open ports

Port	State	Service	Version
21/tcp	OPEN	FTP	ProFTPD 1.3.5
22/tcp	OPEN	SSH	OpenSSH 6.6.1p1 Ubuntu

80/tcp	OPEN	HTTP	Apache httpd 2.4.7
111/tcp	OPEN	RPC	rpcbind 2-4
139/tcp	OPEN	NetBIOS	Samba 3.X-4.X
445/tcp	OPEN	SMB	Samba 3.X-4.X
3306/tcp	OPEN	MySQL	MySQL (unauthorized)
6667/tcp	OPEN	IRC	UnrealIRCd
8080/tcp	OPEN	HTTP	Jetty 8.1.7

SIEM Detection: FTP and SSH connection attempts from 10.0.2.4 were captured in Splunk. The scan revealed 9 open ports, significantly increasing the attack surface. **Remediation:** Disable unnecessary services (IRC, RPC, MySQL) and implement a host-based firewall.

FINDING 2 — SSH Brute Force Attack SEVERITY: CRITICAL

Tool	Hydra
Command	hydra -l vagrant -P rockyou.txt ssh://10.0.2.15 -t 4
Time to Crack	Under 5 seconds
Credentials Found	vagrant / vagrant (default credentials)
Events Generated	1,830+ in one hour

SIEM Detection: Splunk captured over 1,830 authentication events within one hour, with a peak of 200+ events per minute. Log entries showed repeated 'Failed password for vagrant from 10.0.2.4' in sshd, clearly identifying the brute force pattern. The attack succeeded due to default credentials never being changed after provisioning. **Remediation:** Change all default credentials immediately. Implement account lockout after 5 failed attempts using fail2ban.

FINDING 3 — Remote Code Execution via ProFTPD CVE SEVERITY: CRITICAL

Tool	Metasploit Framework 6.4
Module	exploit/unix/ftp/proftpd_modcopy_exec
Vulnerability	ProFTPD 1.3.5 mod_copy — unauthenticated file copy
Result	Remote shell obtained (uid=1000, sudo group membership)
Access Level	Full system compromise

SIEM Detection: Splunk logged FTP sessions from 10.0.2.4, PHP payload copy attempts, and subsequent session activity. The mod_copy module in ProFTPD 1.3.5 allows any unauthenticated client to copy files from any part of the filesystem. A reverse TCP shell was established from 10.0.2.15:33812 to 10.0.2.4:4444, granting full command execution with sudo privileges. **Remediation:** Upgrade ProFTPD to 1.3.6+ or disable the mod_copy module immediately.

Remediation Summary

#	Recommendation	Priority
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1	Change all default credentials — vagrant/vagrant is publicly known	CRITICAL
2	Upgrade ProFTPD to 1.3.6+ or disable mod_copy module	CRITICAL
3	Implement SSH account lockout after 5 failed attempts (fail2ban)	HIGH
4	Close unnecessary ports — MySQL (3306), IRC (6667), RPC (111)	HIGH
5	Deploy IDS/IPS with brute force detection rules	HIGH
6	Configure Splunk alert when failed logins exceed 10 per minute	MEDIUM

Conclusion

All three attack phases were successfully detected by the Splunk SIEM, demonstrating the effectiveness of centralized log monitoring in identifying active threats in real time. The target system exhibited multiple critical vulnerabilities including default credentials, an unpatched remote code execution vulnerability, and an excessive number of exposed network services. These findings underscore the importance of patch management, credential hygiene, and continuous security monitoring. This assessment demonstrates end-to-end SOC analyst workflow: reconnaissance detection, brute force identification, exploit detection, and structured incident documentation.